

Fractions

Mixed numbers to top-heavy fractions first — then add with a common denominator, multiply straight across, and divide with KFC.



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THE TOPIC ON ONE MAP

Fractions

01 — Fraction basics

$\frac{7}{11}$ of 198 kg

£1.50 as a fraction of 60p

lowest terms

half way between

02 — Add and subtract

$4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$

$5\frac{1}{3} - 2\frac{6}{7}$

“as a mixed number”

03 — Multiply and divide

$3\frac{1}{5} \times 2\frac{5}{8}$

$6\frac{3}{4} \div 2\frac{4}{7}$

“show that”

04 — Fraction problems

rent, travel, \$420 left

$\frac{7}{20}$ of the bricks

ratio of students

WHICH MOVE? Every calculation starts the same way — mixed numbers become top-heavy:

ZONE + −

Common denominator

same bottom, then add / subtract the tops

ZONE ×

Straight across

top × top, bottom × bottom (cancel first)

ZONE ÷

KFC

Keep – Flip – Change to ×

01 Fraction Basics

“Of” means multiply. “As a fraction of” means the first amount on top — in the **same units**. Then cancel to lowest terms.

WHEN YOU SEE → YOU DO

“ $\frac{7}{11}$ of 198 kg”

→ $198 \div 11 \times 7$

“£1.50 as a fraction of 60p”

→ same units: $\frac{150}{60} = \frac{5}{2}$

“in its lowest terms / simplest form”

→ divide top and bottom by the HCF

“half way between two fractions”

→ add them, divide by 2

WORKED MODEL • FRACTION OF AN AMOUNT

Work out $\frac{7}{11}$ of 198 kg.

1 $198 \div 11 = 18$

one eleventh

2 18×7

seven elevenths

ANSWER 126 kg

× **TRAP** £1.50 and 60p are different units: $\frac{1.50}{60}$ is wrong. Change both to pence first.

IDEA 01

Practise: Fraction Basics

My questions first, then selected practice.

Q1 **DR ESLAM** Of an amount

CH 1 • TOPIC 5 2 marks

Work out $\frac{3}{8}$ of 240 kg.**SELECTED PRACTICE****Q2** **PRACTICE** Practice • Easy

CH 1 • TOPIC 5 1 mark

3 Work out £1.50 as a fraction of 60p

Circle your answer.

$\frac{2}{5}$	$\frac{1}{4}$	$\frac{4}{1}$	$\frac{5}{2}$
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(1 mark)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 5**1 mark**

4 Circle the fraction that is equivalent to 3.875

$\frac{15}{4}$	$\frac{29}{8}$	$\frac{31}{8}$	$\frac{15}{8}$
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(1 mark)

Q4**PRACTICE**

Practice • Easy

CH 1 • TOPIC 5**2 marks**

5 Write \$24.60 as a fraction of \$2870.

Give your answer in its lowest terms.

(2 marks)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 5 1 mark

- 11 Circle the number half way between $\frac{7}{12}$ and $\frac{3}{4}$

$\frac{7}{32}$	$\frac{5}{8}$	$\frac{2}{3}$	$\frac{1}{2}$
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(1 mark)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 5 3 marks

- 13** A unit fraction has a numerator equal to 1, for example $\frac{1}{3}$, $\frac{1}{7}$ and $\frac{1}{25}$.

Unit fractions can be written as the sum of two different unit fractions, for example

$$\frac{1}{2} = \frac{1}{3} + \frac{1}{6}$$

Write each of the following unit fractions as the sum of two **different** unit fractions.

$$\frac{1}{4} = \frac{1}{\square} + \frac{1}{\square}$$

$$\frac{1}{5} = \frac{1}{\square} + \frac{1}{\square}$$

$$\frac{1}{6} = \frac{1}{\square} + \frac{1}{\square}$$

(3 marks)

Q7**PRACTICE**

Practice • Medium

CH 1 • TOPIC 5 1 mark

17 Giulio's reaction times are measured in two games.

In the first game his reaction time is $\frac{1}{3}$ of a second.

In the second game his reaction time is $\frac{1}{8}$ of a second.

Find the difference between the two reaction times.

..... S

(1 mark)

02 Add and Subtract

Top-heavy first, then one common denominator. Multiply each top by what its bottom was multiplied by.

WHEN YOU SEE → YOU DO

“Show that $4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$ ”

→ write both top-heavy, common bottom, convert back

“without a calculator ... simplest form”

→ show every line; cancel at the end

“as a mixed number”

→ divide: top ÷ bottom, remainder over the bottom

subtraction where the first fraction is smaller

→ top-heavy avoids the borrowing problem

Top-heavy

Common denominator

Add / subtract tops

Back to mixed

WORKED MODEL • JUNE 2022 P2H

Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$.

1 $\frac{16}{3} - \frac{20}{7}$

top-heavy

2 $\frac{112}{21} - \frac{60}{21}$

common denominator 21

3 $\frac{52}{21}$

subtract the tops

ANSWER $2\frac{10}{21}$

× **TRAP** Adding the bottoms: $\frac{2}{3} + \frac{4}{5} \neq \frac{6}{8}$. Only the tops add, once the bottoms are the same.

IDEA 02

Practise: Add and Subtract

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM Add mixed numbers

CH 1 • TOPIC 5 3 marks

Show that $2\frac{1}{3} + 1\frac{3}{4} = 4\frac{1}{12}$.

Q9 DR ESLAM Subtract

CH 1 • TOPIC 5 3 marks

Work out $3\frac{2}{5} - 1\frac{3}{4}$. Give your answer as a mixed number.

SELECTED PRACTICE

Q10

PRACTICE

Practice • Easy

CH 1 • TOPIC 5 3 marks

1 (a) Show that $\frac{1}{7} \times \frac{2}{3}$ is equal to $\frac{2}{21}$

(1 mark)

(b) Show that $\frac{3}{5} - \frac{1}{3}$ is equal to $\frac{4}{15}$

(2 marks)

Q11

PRACTICE

Practice • Medium

CH 1 • TOPIC 5 3 marks

16 Without using a calculator, work out $\frac{5}{6} + \frac{2}{3}$.

You must show all your working and give your answer as a mixed number in its simplest form.

(3 marks)

Q12**PRACTICE**

Practice · Hard

CH 1 • TOPIC 5**3 marks**

4 Show that $5\frac{2}{3} - 2\frac{3}{4} = 2\frac{11}{12}$

(3 marks)**Q13****PRACTICE**

Practice · Hard

CH 1 • TOPIC 5**3 marks**

15 **Without using a calculator**, workout $3\frac{5}{8} - 1\frac{2}{3}$.

You must show all your working and give your answer as a mixed number in its simplest form.

(3 marks)

PAST PAPERS

Q14 **PAST PAPER** Edexcel 4MA1 • January 2020 • Paper 2H • Q3**CH 1 • TOPIC 5** **3 marks**

3 Show that $4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$

(Total for Question 3 is 3 marks)

Q15 **PAST PAPER** Edexcel 4MA1 • June 2022 • Paper 1HR • Q6**CH 1 • TOPIC 5** **3 marks**

6 Show that $2\frac{2}{3} + 3\frac{3}{4} = 6\frac{5}{12}$

(Total for Question 6 is 3 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q4

CH 1 • TOPIC 5 3 marks

4 Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

(Total for Question 4 is 3 marks)

Q17**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q3

CH 1 • TOPIC 5 3 marks

3 Show that $7\frac{1}{3} - 3\frac{4}{7} = 3\frac{16}{21}$

(Total for Question 3 is 3 marks)

Q18**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q5

CH 1 • TOPIC 5 3 marks

5 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

(Total for Question 5 is 3 marks)

03 Multiply and Divide

Top-heavy first. Multiply: straight across (cancel first to keep numbers small). Divide: **KFC** — Keep the first, Flip the second, Change $\div$ to $\times$.

WHEN YOU SEE → YOU DO

" $3\frac{1}{5} \times 2\frac{5}{8}$ "

→ $\frac{16}{5} \times \frac{21}{8}$, cancel 16 with 8

" $\div$ "

→ Keep – Flip – Change

"You must write down all the stages"

→ top-heavy line, KFC line, answer line

a whole number in the answer

→ $\frac{56}{7} = 8$

Top-heavy

KFC if $\div$

Cancel

Multiply across

Mixed number

WORKED MODEL • JANUARY 2022 P2H

Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$.

1 $\frac{27}{4} \div \frac{18}{7}$

top-heavy

2 $\frac{27}{4} \times \frac{7}{18}$

KFC

3 $\frac{3 \times 7}{4 \times 2} = \frac{21}{8}$

cancel 27 and 18 by 9

ANSWER $2\frac{5}{8}$

× **TRAP** $3\frac{1}{5} \times 2\frac{5}{8}$ is not $6 + \frac{1}{5} \times \frac{5}{8}$. Never multiply the whole numbers and the fractions separately.

IDEA 03

Practise: Multiply and Divide

My questions first, then selected practice, then real past papers.

Q19 DR ESLAM KFC

CH 1 • TOPIC 5 3 marks

Show that $2\frac{1}{4} \div 1\frac{1}{2} = 1\frac{1}{2}$.

SELECTED PRACTICE**Q20** PRACTICE Practice • Medium

CH 1 • TOPIC 5 6 marks

1 (a) Show that $1\frac{1}{5} \times 2\frac{1}{3} = 2\frac{4}{5}$

(3 marks)

(b) Show that $2\frac{7}{15} - 1\frac{2}{3} = \frac{4}{5}$

(3 marks)

Q21 PRACTICE Practice • Medium

CH 1 • TOPIC 5 2 marks

20 Without using a calculator, work out $\frac{1}{4} \div \frac{2}{3}$.

You must show all your working and give your answer as a fraction.

(2 marks)

Q22 PRACTICE Practice • Hard

CH 1 • TOPIC 5 2 marks

14 Without using a calculator, work out $\frac{5}{16} \times 1\frac{1}{7}$.

You must show all your working and give your answer as a fraction in its simplest form.

(2 marks)

Q23**PRACTICE**

Practice • Hard

CH 1 • TOPIC 5**3 marks**

20 Without using a calculator, work out $\frac{2}{3} \div 1\frac{1}{5}$.

You must show all your working and give your answer as a fraction in its simplest form.

(3 marks)

PAST PAPERS

Q24**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1HR • Q2

CH 1 • TOPIC 5**3 marks**

2 Show that $3\frac{1}{5} \times 2\frac{5}{8} = 8\frac{2}{5}$

(Total for Question 2 is 3 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q1

CH 1 • TOPIC 5 3 marks

You must write down all the stages in your working.

1 Show that $3\frac{3}{4} \times \frac{7}{9} = 2\frac{11}{12}$

(Total for Question 1 is 3 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q1

CH 1 • TOPIC 5 3 marks

You must write down all the stages in your working.

1 Show that $3\frac{1}{5} \times 1\frac{5}{6} = 5\frac{13}{15}$

(Total for Question 1 is 3 marks)

Q27 **PAST PAPER** Edexcel 4MA1 • May 2021 • Paper 2H • Q2**CH 1 • TOPIC 5** **3 marks**

2 Show that $2\frac{4}{7} \div 1\frac{1}{8} = 2\frac{2}{7}$

(Total for Question 2 is 3 marks)

Q28 **PAST PAPER** Edexcel 4MA1 • January 2022 • Paper 2H • Q2**CH 1 • TOPIC 5** **3 marks**

2 Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$

(Total for Question 2 is 3 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q1

CH 1 • TOPIC 5 3 marks

You must write down all the stages in your working.

1 Show that $4\frac{2}{3} \div 1\frac{1}{5} = 3\frac{8}{9}$

(Total for Question 1 is 3 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q6

CH 1 • TOPIC 5 3 marks

6 Show that $3\frac{3}{7} \div 2\frac{2}{3} = 1\frac{2}{7}$

(Total for Question 6 is 3 marks)

Q31 **PAST PAPER** Edexcel 4MA1 • June 2024 • Paper 2H • Q4**CH 1 • TOPIC 5** **3 marks**

4 Show that $2\frac{4}{7} \times 3\frac{1}{9} = 8$

(Total for Question 4 is 3 marks)

Q32 **PAST PAPER** Edexcel 4MA1 • May 2024 • Paper 1HR • Q4**CH 1 • TOPIC 5** **3 marks**

4 Show that $2\frac{1}{3} \div 5\frac{1}{4} = \frac{4}{9}$

(Total for Question 4 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q1

CH 1 • TOPIC 5 3 marks

You must write down all the stages in your working.

1 Show that $1\frac{5}{7} \times 2\frac{3}{16} = 3\frac{3}{4}$

(Total for Question 1 is 3 marks)

Q34**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q3

CH 1 • TOPIC 5 3 marks

3 Show that $2\frac{1}{4} \times 1\frac{5}{7} = 3\frac{6}{7}$

(Total for Question 3 is 3 marks)

04 Fraction Problems

Fractions of a whole: add the fractions you know, the rest is 1 – that. Then match the rest to the amount you are given.

WHEN YOU SEE → YOU DO

“she spends $\frac{1}{3}$... $\frac{1}{5}$... and \$420”

→ \$420 is the fraction left: $1 - \frac{1}{3} - \frac{1}{5}$

“ $\frac{7}{20}$ of the bricks are yellow”

→ the others are $\frac{13}{20}$ – that matches the 52 you know

“ $\frac{2}{3}$ of the students ... $\frac{2}{5}$ of those”

→ “of those” → multiply

“ratio of ... to ...”

→ find both fractions, then write as a ratio

WORKED MODEL • NOVEMBER 2025 P2H

9 red, 43 blue, and yellow bricks; $\frac{7}{20}$ are yellow; each brick 35 g. Weight of the yellow bricks?

1 red + blue = 52 = $\frac{13}{20}$ of the bricks

the known part

2 $\frac{1}{20} = 4$ bricks, yellow = $7 \times 4 = 28$

one part

ANSWER $28 \times 35 = 980$ g

IDEA 04

Practise: Fraction Problems

My questions first, then selected practice, then real past papers.

Q35 DR ESLAM The rest

CH 1 • TOPIC 5 3 marks

Sam spends $\frac{1}{4}$ of his money on food and $\frac{2}{5}$ on rent. He has \$70 left. How much did he start with?

SELECTED PRACTICE**Q36** PRACTICE Practice • Medium

CH 1 • TOPIC 5 3 marks

12 In a school, $\frac{2}{3}$ of the students study a language.

Of those students who study a language, $\frac{2}{5}$ study Spanish.

Find the ratio of students who study Spanish to students who do not study Spanish.

(3 marks)

Q37**PRACTICE**

Practice • Medium

CH 1 • TOPIC 5 3 marks

- 21** Collette has \$136 and spends half of her money on clothes and $\frac{1}{5}$ of her money on books.

Calculate the amount she has left.

\$

(3 marks)

Q38**PRACTICE**

Practice • Hard

CH 1 • TOPIC 5 4 marks

5 Each month Edna spends all her income on rent, on travel and on other living expenses.

She spends $\frac{1}{3}$ of her income on rent.

She spends $\frac{1}{5}$ of her income on travel.

She spends \$420 of her income on other living expenses.

Work out her income each month.

(4 marks)

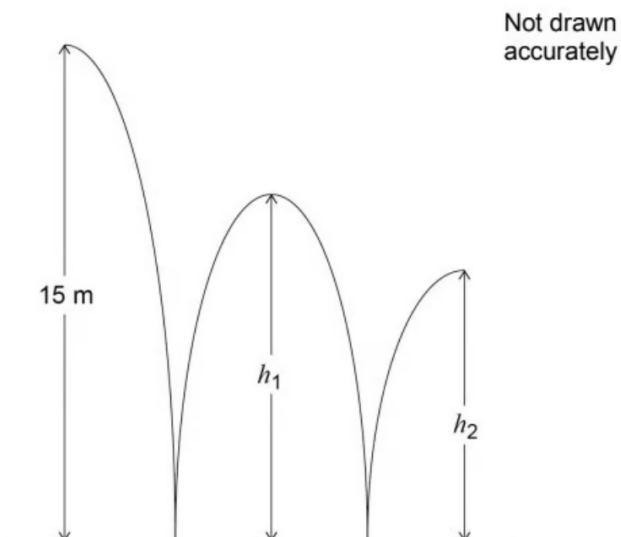
Q39**PRACTICE**

Practice • Hard

CH 1 • TOPIC 5 4 marks

6 (a) A ball is thrown from a height of 15 metres.

It bounces to height h_1 , then to height h_2 as shown.



h_1 is three quarters of the original height.

Jack expects h_2 to be three quarters of h_1

Work out the value of h_2 that he expects.

.....metres

(2 marks)

(b) In fact, h_2 is two thirds of h_1

How does this affect the answer to part (a)? Tick a box.

☐ The ball bounced higher than he expected

☐ The ball bounced lower than he expected

Show working to support your answer.

(2 marks)

PAST PAPERS

Q40**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q3

CH 1 • TOPIC 5 4 marks

3 A box contains only

9 red bricks
43 blue bricks
and some yellow bricks

$\frac{7}{20}$ of the bricks are yellow bricks.

Each brick weighs 35 grams.

Work out the total weight of the yellow bricks.

..... grams

(Total for Question 3 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE + –
same bottom
then the tops

ZONE ×
across
cancel first

ZONE ÷
KFC
Keep–Flip–Change

- | | |
|-----------------------------|---|
| 01 Basics | “Of” = $\times$. “As a fraction of”: same units, then cancel. |
| 02 Add / subtract | Top-heavy $\rightarrow$ common denominator $\rightarrow$ tops $\rightarrow$ mixed number. |
| 03 Multiply / divide | Top-heavy; $\times$ across; $\div$ KFC; cancel first. |
| 04 Problems | Known part = 1 – fractions; find one part, then the rest. |

BOTTOM LINE Mixed numbers $\rightarrow$ top-heavy, always $\rightarrow$ then the right move $\rightarrow$ back to a mixed number.

Answers

Fractions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • FRACTION BASICS

- | | | |
|-----------|-------------------------|--|
| Q1 | Dr Eslam • Of an amount | $240 \div 8 \times 3 = 90 \text{ kg}$ |
| Q2 | Practice • Easy | $\frac{5}{2}$ |
| Q3 | Practice • Easy | $\frac{31}{8}$ |
| Q4 | Practice • Easy | $\frac{3}{350}$ |
| Q5 | Practice • Medium | $\frac{2}{3}$ |
| Q6 | Practice • Medium | $\frac{1}{4} = \frac{1}{5} + \frac{1}{20}, \frac{1}{5} = \frac{1}{6} + \frac{1}{30}, \frac{1}{6} = \frac{1}{7} + \frac{1}{42}$ |
| Q7 | Practice • Medium | $\frac{5}{24} \text{ s}$ |

IDEA 02 • ADD AND SUBTRACT

- Q8** Dr Eslam • Add mixed numbers $\frac{7}{3} + \frac{7}{4} = \frac{28}{12} + \frac{21}{12} = \frac{49}{12}$
- Q9** Dr Eslam • Subtract $\frac{68}{20} - \frac{35}{20} = \frac{33}{20} = 1\frac{13}{20}$
- Q10** Practice • Easy (a) $\frac{1 \times 2}{7 \times 3}$ (b) $\frac{9}{15} - \frac{5}{15}$
- Q11** Practice • Medium $1\frac{1}{2}$
- Q12** Practice • Hard $\frac{68}{12} - \frac{33}{12} = \frac{35}{12}$
- Q13** Practice • Hard $1\frac{23}{24}$
- Q14** Edexcel 4MA1 • January 2020 • Paper 2H • Q3 $\frac{70}{15} + \frac{57}{15} = \frac{127}{15}$
- Q15** Edexcel 4MA1 • June 2022 • Paper 1HR • Q6 $\frac{32}{12} + \frac{45}{12} = \frac{77}{12}$
- Q16** Edexcel 4MA1 • June 2022 • Paper 2H • Q4 $\frac{112}{21} - \frac{60}{21} = \frac{52}{21}$
- Q17** Edexcel 4MA1 • June 2025 • Paper 2H • Q3 $\frac{154}{21} - \frac{75}{21} = \frac{79}{21}$
- Q18** Edexcel 4MA1 • November 2025 • Paper 1H • Q5 $\frac{78}{21} + \frac{35}{21} = \frac{113}{21}$

IDEA 03 • MULTIPLY AND DIVIDE

Q19	Dr Eslam • KFC	$\frac{9}{4} \times \frac{2}{3} = \frac{18}{12} = \frac{3}{2}$
Q20	Practice • Medium	(a) $\frac{6}{5} \times \frac{7}{3} = \frac{14}{5}$ (b) $\frac{37}{15} - \frac{25}{15} = \frac{12}{15}$
Q21	Practice • Medium	$\frac{3}{8}$
Q22	Practice • Hard	$\frac{5}{14}$
Q23	Practice • Hard	$\frac{5}{9}$
Q24	Edexcel 4MA1 • January 2020 • Paper 1HR • Q2	$\frac{16}{5} \times \frac{21}{8} = \frac{42}{5}$
Q25	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q1	$\frac{15}{4} \times \frac{7}{9} = \frac{35}{12}$
Q26	Edexcel 4MA1 • January 2021 • Paper 1HR • Q1	$\frac{16}{5} \times \frac{11}{6} = \frac{88}{15}$
Q27	Edexcel 4MA1 • May 2021 • Paper 2H • Q2	$\frac{18}{7} \times \frac{8}{9} = \frac{16}{7}$
Q28	Edexcel 4MA1 • January 2022 • Paper 2H • Q2	$\frac{27}{4} \times \frac{7}{18} = \frac{21}{8}$
Q29	Edexcel 4MA1 • June 2023 • Paper 2H • Q1	$\frac{14}{3} \times \frac{5}{6} = \frac{35}{9}$
Q30	Edexcel 4MA1 • November 2023 • Paper 2H • Q6	$\frac{24}{7} \times \frac{3}{8} = \frac{9}{7}$
Q31	Edexcel 4MA1 • June 2024 • Paper 2H • Q4	$\frac{18}{7} \times \frac{28}{9} = 8$
Q32	Edexcel 4MA1 • May 2024 • Paper 1HR • Q4	$\frac{7}{3} \times \frac{4}{21} = \frac{4}{9}$
Q33	Edexcel 4MA1 • November 2024 • Paper 2H • Q1	$\frac{12}{7} \times \frac{35}{16} = \frac{15}{4}$
Q34	Edexcel 4MA1 • June 2025 • Paper 1HR • Q3	$\frac{9}{4} \times \frac{12}{7} = \frac{27}{7}$

IDEA 04 • FRACTION PROBLEMS

Q35 Dr Eslam • The rest

$$\text{left} = \frac{7}{20}: 70 \div 7 \times 20 = \$200$$

Q36 Practice • Medium

$$\frac{4}{15} : \frac{11}{15} = 4 : 11$$

Q37 Practice • Medium

\$40.80

Q38 Practice • Hard

$$\$420 \text{ is } \frac{7}{15}: \$900$$

Q39 Practice • Hard

(a) $8\frac{7}{16}$ m (b) $h_2 = 7.5$ m < $8\frac{7}{16}$: lower than expected

Q40 Edexcel 4MA1 • November 2025
• Paper 2H • Q3

80 bricks, 28 yellow: 980 g